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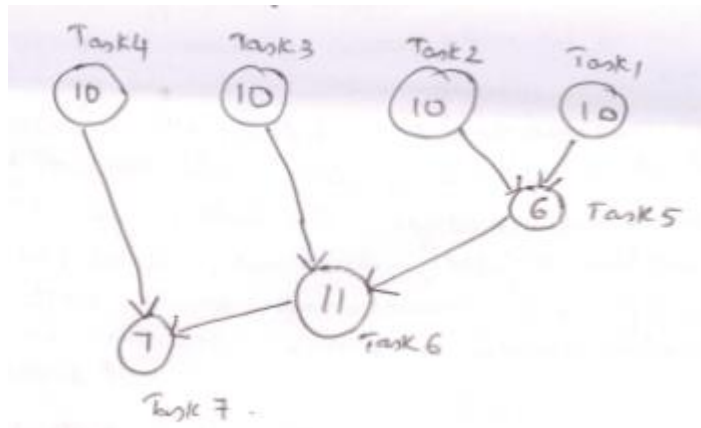
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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations Feb/Mar 2023
Computer Science & Engineering
PARALLEL ARCHITECTURE & DISTRIBUTED
PROGRAMMING

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Give the Flynn's taxonomy of Parallel Architectures	02
	1.2	Write VMIPS code to add two vectors	02
	1.3	Define Chaining	01
	1.4	 Fig 1.4 For the graph in Fig 1.4 calculate the critical path length and average degree of concurrency.	
	1.5	Define mapping	01
	1.6	Give the syntax for MPI_Send and MPI_Recv system calls.	02
	1.7	Give the basic structure of MPI Program	02
	1.8	According to amdahl's law give two goals that developer has to keep in mind when parallelizing application	02
	1.9	Give any two reasons to say memory system is coherent	02
	1.10	List the methods/types to implement snooping protocol	02
	1.11	Differentiate between fine grained and coarse grained decomposition.	02

PART-B

2	a Suppose we have an application running on a 32 processor multiprocessor, which has a 200ns time to handle reference to a remote memory: For this application, assume that all the references except those involving communication hit in the local memory hierarchy. Which is slightly optimistic? Processors are stalled on remote request, and the processor clock rate is 2GHz. If the base CPI (assuming that all references hit in the cache) is 0.5, how much faster is multiprocessor if there is no communication versus if 0.2% of the instructions involve a remote communication reference? b Illustrate directory based Protocol with state transition diagram c Explain atomic exchange in synchronization	06 08 02
3	a How do you use multiple functional units to improve performance of a single vector add instruction $C=A+B$. b Consider the loop for(i=0; i<100; i=i+1) { A[i+1] = A[i] + C[i]; /*S1*/ B[i+1] = B[i] + A [i+1]; /*S2*/ } Assume that A, B, C are distinct, non overlapping arrays. (In Practice, arrays may sometimes be same or may overlap. Because arrays may be passed as parameter to procedure that includes this loop determining whether arrays overlap or are identical often requires sophisticated, interprocedural analysis of a program). What are the data dependences among statements S1 and S2 in the loop? c Show the code for MIPS and VMIPS for DAXPY loop. Assume that starting address of X and Y are in Rx and Ry respectively. OR	06 04 06
4	a Use GCD test to determine whether dependencies exist in following loop: for(i=0; i<100; i=i+1) { x[2*i+3] = x[2*i]*5.0; } } b Show how the following sequence lays out in registers, assuming single copy of each vector functional unit: LV V₁,Rx MULVS.D V₂,V₁,F₀ LV V₃,Ry ADDVV.D V₄,V₂,V₃ SV V₄,Ry How many cycles will this vector sequence take? How many cycles per FLOP are needed ignore vector instruction overhead? c Illustrate VMIPS architecture with a neat diagram.	04 04 04 08

5	a	Taking an example of 15-puzzle problem illustrate exploratory decomposition.	08
	b	Explain the following characteristics of tasks: i) Task generation ii) Task size iii) Knowledge of task size iv) Size of data associated with tasks.	08
OR			
6	a	Illustrate any 3 parallel algorithm models in detail.	09
	b	Taking an example of Quicksort show how recursive decomposition works. Write the steps in detail.	07
7	a	Illustrate the building blocks send and receive operations.	06
	b	Give the different collective communication and computation operations used in MPI.	10
8	a	Illustrate the 3 steps to accelerate with openACC	04
	b	Illustrate various forms of openACC Parallelism	06
	c	Briefly describe how pipeline is implemented across multiple devices in openACC?	06